Mostafa Alrahal

Applied AI Engineer | LLM, RAG & Agentic AI Systems

US Permanent Resident | +1 (724) 351-7270 | mostafa.kha.alrahal@gmail.com

Pittsburgh, PA | LinkedIn | Portfolio | GitHub

Applied AI Engineer with hands-on experience building LLM applications, RAG pipelines, and agentic AI systems using Python, LangGraph, LangChain, Google ADK, and FastAPI. Experienced in document ingestion, retrieval architectures, prompt engineering, tool calling, and multi-agent orchestration. Contributed to Deloitte's AI Tax Assistant by developing RAG and document-processing workflows over thousands of proprietary legal documents, and built multi-tenant conversational AI systems for customer-specific use cases. U.S. Permanent Resident with a WES-verified U.S. bachelor's-equivalent degree in Information Technology Engineering.

Work Experience

Maly Tech Ltd | Gen AI Engineer | Dubai, UAE **Aug 2024 – May 2026**

- Contributed to the development and deployment of **Deloitte's AI Tax Assistant**, implementing RAG, LangGraph, and prompt engineering to ingest, index, and retrieve information from thousands of proprietary legal documents for MENA tax guidance.
- Built **AI4ALL, a multi-tenant AI platform** for rapidly deploying and demonstrating customer-specific AI agents and prototypes, using Python, Google ADK, agent orchestration, tool calling, and prompt engineering; reduced the time required to deliver new AI prototypes from days or weeks to hours.
- Developed **Luna, a conversational AI financial assistant,** leveraging RAG, prompt engineering (Few-Shot, ReAct), and context-injection to deliver safe, personalized answers to financial inquiries, reducing routine customer support tickets by 60%.

Freelance | Software Developer **Oct 2023 – Aug 2024**

- Developed and maintained production web applications using **Angular, Laravel, and REST APIs**, building reusable UI components and integrating backend services to support feature delivery.

Projects

Multi-Agent Travel Planner **2026**

- Built a LangGraph **multi-agent travel planner** that orchestrates specialized agents and dynamically provisions destination-specific RAG knowledge bases when information is unavailable from the indexed knowledge base.

Document Processing Pipeline **2026**

- Designed and implemented a **multi-tenant data ingestion service** for use with RAG system, automating the parsing and vectorization of PDFs, DOCX, PPTX, TXT, and MD files using Docling, LlamaParse, and Jina's late chunking, which accelerated the creation of new agentic RAG applications.

Graduation Project | Fruitful Tree Count Management **2024**

- Fine-tuned custom **deep learning model** and developed **OpenCV computer-vision algorithms** to automate fruit-tree counting and land-cover detection from satellite imagery, achieving >90% accuracy.

Academic AI Projects **2021 – 2024**

- Built applied NLP, computer-vision, deep-learning, and machine-learning projects, including hand-gesture recognition, text classification, and income prediction using OpenCV and Scikit-learn, PyTorch.

Education

Damascus University | BS in Information Technology Engineering **2025**

- **Specialization:** Artificial Intelligence & Natural Language Processing.
- **GPA: 3.78/4.0** (WES Verified U.S. Bachelor's Equivalent).

Skills

- **AI & ML:** Generative AI, Multi-Agent Orchestration, Prompt Engineering, Tool Calling, Knowledge Graphs, Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Natural Language Processing (NLP), Computer Vision (CV).
- **Frameworks & Libraries:** LangGraph, LangChain, Google ADK, PyTorch, Scikit-learn, OpenCV, Docling.
- **Tools:** Python, Java, Typescript, FastAPI, Laravel, Angular, PostgreSQL, MongoDB, Vector DBs (pgvector, Atlas), Graph DBs (Neo4j), Redis, GCP, Docker, Git, RESTful APIs, Streaming APIs (SSE, WebSockets).